

2022 Yamaha YZ250F Track-Day Pack

Printable add-on for practice / race day. Works with the free workshop app (Jobs · Specs · Maps · Kits) — the app stays free; this pack is the garage printout.

Built for: 2019–2023 YZ250F generation (verify your year against the Owner's Service Manual).

Air setup note (Kyle's bike): Twin Air Powerflow **152222C** — use Powerflow filters with that kit; stock cage uses OEM **B2W-14451-01**.

Pricing: Launch **\$9.99** → Standard **\$12.99**. Free workshop app stays ungated — this PDF is the printable garage add-on only. No proprietary .map file dumps.

DISCLAIMER — READ BEFORE YOU WRENCH

Not Yamaha OEM warranty advice. Torque values below are from OEM-lineage research for this generation — VERIFY every critical fastener in your year-specific Owner's Service Manual before you wrench. Race / closed-course use. Power Tuner grids are relative trims; enter them yourself in the official app. Wrong axes or bad transcription can hurt performance or the engine. If unsure, reset maps to all zeros (stock).

1. Pre-ride checklist

Check each box before you leave the truck.

Fluids & filter

- ☐ Engine oil level between marks (warm bike preferred; don't overfill — wet clutch)
- ☐ Coolant level / no leaks at water pump & hoses
- ☐ Air filter clean & oiled (**152222C** Powerflow element, or OEM if stock cage)
- ☐ Airbox sealed (no loose screws / crushed foam)

Drivetrain

- ☐ Chain slack in spec; lube after wash / dusty moto
- ☐ Sprocket bolts / lock tabs look right
- ☐ Rear wheel alignment marks equal both sides

Controls & brakes

- ☐ Front / rear brake lever & pedal firm (no air feel)
- ☐ Throttle free play OK; snaps shut
- ☐ Clutch lever free play OK
- ☐ Kill switch works

Chassis & fasteners (quick pass)

- ☐ Axle nuts / pinch bolts snug (see torque sheet — **VERIFY**)
- ☐ Bar clamps / triple clamps (if you moved bars)
- ☐ Spokes (quick hand check)
- ☐ Tire pressure set for today's track (write PSI below)

Today's PSI: Front _____ Rear _____

Track / date: _____

Map 1 / Map 2 loaded: _____ / _____

Page 1 disclaimer (print): Not Yamaha OEM warranty advice. Torque values below are from OEM-lineage research for this generation — VERIFY every critical fastener in your year-specific Owner's Service Manual before you wrench. Race / closed-course use. Power Tuner grids are relative trims; enter them yourself in the official app. Wrong axes or bad transcription can hurt performance or the engine. If unsure, reset maps to all zeros (stock).

2. Post-ride checklist

Do this before the bike sits dirty overnight.

- ☐ Hose off mud (avoid blasting bearings / airbox intake)
- ☐ Air filter out → clean / oil / dry → reinstall
- ☐ Chain clean + lube
- ☐ Quick look for oil / coolant weep
- ☐ Note anything that felt off (fork harsh, brake fade, bog, overheating)
- ☐ Log hours / motos on the blank log page
- ☐ Phone charged + Power Tuner maps still assigned how you want for next day

Notes: _____

3. Torque quick-ref (critical only)

VERIFY against 2022 YZ250F OSM. Prefer OEM single values over blog ranges. MotoSport-style blog axle ranges have conflicted with OEM — **do not use those**.

Fastener	Target	Notes
Engine oil drain bolt	20 Nm (14 ft·lbf)	New crush washer every change
Oil filter cover bolts	10 Nm (7.2 ft·lbf)	M6 into aluminum — don't overtighten; new O-ring 93210-45708
Spark plug (NGK LMAR9E-J)	13 Nm (9.4 ft·lbf)	Hand-thread first
Air filter fitting (older OSM lineage)	~2 Nm	Easy to crush foam/cage — light touch; confirm 2022 hardware
Front axle nut (OSM lineage)	115 Nm (83 ft·lbf)	Then pinch bolts — VERIFY 2022 chart
Front axle pinch	21 Nm (15 ft·lbf)	After axle seated
Rear axle nut (OSM lineage)	135 Nm (98 ft·lbf)	Align chain first — VERIFY 2022 chart
Chain adjuster locknut	21 Nm (15 ft·lbf)	Equal marks both sides
Front caliper bolts	28 Nm (20 ft·lbf)	Medium confidence — VERIFY
Brake banjo (F/R)	30 Nm (22 ft·lbf)	New crush washers both sides
Bar clamp / triples	See OSM	Front bolts first on bars; VERIFY ranges

Oil change reminder: Yamalube FS **10W-40** · filter **5D3-13440-09-00** · ~1.05 L with filter · warm drain · idle · recheck level.

4. Power Tuner maps — how to load (no .map dumps)

Your bike holds **two maps**. Bar switch flips between them (**throttle closed**). Stock often ships the **same map in both slots** — Map 2 only gets useful after you upload a second map.

Load steps

1. Install free **Yamaha Power Tuner** (iOS / Android).
2. Phone Wi-Fi → bike **CCU** (password under **left side panel**).
3. Map List → + → name the map.
4. Set **POSITION / RPM axes** to match the chart you're copying (wrong scale makes a good map feel dead).
5. Enter **FI** (fuel) and **IG** (ignition) cells from a verified card / screenshot.
6. Upload (up arrow) to **Map 1** or **Map 2**.
7. Bar: **light off = Map 1, blue = Map 2**.

Share reality: Phones can share proprietary .map files in-app. This pack does **not** include binary map files. Copy numbers → enter in app → upload.

Starter pair for track day (Kyle's verified set)

Slot	Map	When
Map 1	Smooth Linear (Kyle)	Default / learning / slippery — richer top FI, IG retarded for smoother delivery
Map 2	Map10 (Kyle) Hard / Rich FI or Loam / Deep Sand (Kyle)	Punch / rich FI, or soft ripped / deep sand

Map11 = same numbers as Smooth Linear (alias — don't treat as a third tune).

Live FI/IG grids + screenshots also live in the free app **Maps** tab — print this pack for steps/logs; re-check the app if numbers may have updated.

Still empty (on purpose): Hard Pack · Mud / Slippery · Cool Morning · Hot Afternoon — wait for verified screenshots; no fakes.

Kyle's axes (match these in Power Tuner before typing cells)

- **FI POSITION TH:** 20 / 40 / 60 / 80 · **RPM:** 5k / 7.5k / 10k / 12k
- **IG POSITION TH:** 15 / 30 / 50 / 70 · **RPM:** 5k / 7.5k / 10k / 12k

Grids below print TH ascending left→right (20→80 / 15→70) to match the blank sheet. Power Tuner may show high-TH on the left — match by TH% label, not by column index.

Quick map cheat sheet

- **Smooth Linear (Kyle)** — verified Map1 baseline; Map11 is an alias of this.
- **Map10 (Kyle)** — verified hard / rich FI Map2 option.
- **Loam / Deep Sand (Kyle)** — verified for soft / deep sand.
- **Keefer Free Feeling / Top End-Rev** — verified from Keefer charts on the app (freer chassis / mid-top). Copy from app — not reprinted as filled grids here.
- **Yamaha Exciting Power / TP Map 2019** — verified profile candidates on the app; confirm import year in Power Tuner.
- **Hard Pack · Mud · Cool · Hot** — empty until Kyle's screenshots land.

Clean air filter assumed (**152222C**). Clogged foam changes how any map feels.

5. Verified FI / IG grids (Kyle screenshots)

Only maps with real numbers from maps.json are filled. Empty slots stay blank. No proprietary .map dumps.

Smooth Linear (Kyle) VERIFIED

Suggested slot: Map 1 · Default / learning / slippery — richer top FI, IG retarded for smoother delivery

Verified from Kyle Power Tuner screenshot. Map11 is an alias of these exact numbers — do not treat Map11 as a third tune.

Fuel injection (FI) — relative trims

POSITION TH: 20 / 40 / 60 / 80 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	20%	40%	60%	80%
5000	3	3	3	2
7500	3	3	2	2
10000	3	2	1	1
12000	3	1	1	0

Ignition (IG) — relative trims

POSITION TH: 15 / 30 / 50 / 70 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	15%	30%	50%	70%
5000	-4	-3	-3	-2
7500	-3	-2	-2	-1
10000	-3	-2	0	0
12000	-2	-1	0	0

Map11 (Kyle) = Smooth Linear alias VERIFIED

Suggested slot: Map 1 (same as Smooth Linear) · Same FI/IG as Smooth Linear — alternate save name only

Identical cells to Smooth Linear. Use Smooth Linear when copying; Map11 is listed so you don't invent a third map.

Fuel injection (FI) — relative trims

POSITION TH: 20 / 40 / 60 / 80 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	20%	40%	60%	80%
5000	3	3	3	2
7500	3	3	2	2
10000	3	2	1	1
12000	3	1	1	0

Ignition (IG) — relative trims

POSITION TH: 15 / 30 / 50 / 70 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	15%	30%	50%	70%
5000	-4	-3	-3	-2
7500	-3	-2	-2	-1
10000	-3	-2	0	0
12000	-2	-1	0	0

Map10 (Kyle) — Hard / Rich FI VERIFIED

Suggested slot: Map 2 · More fuel hit with stock ignition timing (IG zeros)

Verified from Kyle Power Tuner screenshot “Map10”. FI rich mid-top; IG flat zeros.

Fuel injection (FI) — relative trims

POSITION TH: 20 / 40 / 60 / 80 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	20%	40%	60%	80%
5000	6	6	5	4
7500	5	5	4	3
10000	4	4	3	2
12000	3	2	1	1

Ignition (IG) — relative trims

POSITION TH: 15 / 30 / 50 / 70 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	15%	30%	50%	70%
5000	0	0	0	0
7500	0	0	0	0
10000	0	0	0	0
12000	0	0	0	0

Loam / Deep Sand (Kyle) VERIFIED

Suggested slot: Map 2 · Loam or deep sand needing more fuel and softer ignition

Verified from Kyle Power Tuner screenshot. FI heavily enriched at mid-high throttle; IG retarded across the board.

Fuel injection (FI) — relative trims

POSITION TH: 20 / 40 / 60 / 80 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	20%	40%	60%	80%
5000	4	4	4	4
7500	4	4	4	4
10000	3	3	2	2
12000	3	3	2	1

Ignition (IG) — relative trims

POSITION TH: 15 / 30 / 50 / 70 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	15%	30%	50%	70%
5000	-4	-4	-3	-3
7500	-4	-3	-3	-2
10000	-3	-2	-2	-1
12000	-3	-2	-2	-1

Hard Pack EMPTY — await Kyle screenshot

Suggested slot: Map 2 · Firm, slick hardpack — tractable drive, less wheelspin. Wait for verified screenshot.

EMPTY on purpose — blank grid with Kyle’s axes only. No invented numbers. Copy from a verified screenshot later.

Fuel injection (FI) — relative trims

POSITION TH: 20 / 40 / 60 / 80 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	20%	40%	60%	80%
5000				
7500				
10000				
12000				

Ignition (IG) — relative trims

POSITION TH: 15 / 30 / 50 / 70 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	15%	30%	50%	70%
5000				
7500				
10000				
12000				

Mud / Slippery EMPTY — await Kyle screenshot

Suggested slot: Map 1 or Map 2 · Wet, slick, muddy — softer delivery to cut spin. Empty until Kyle's screenshot lands.

EMPTY on purpose — blank grid with Kyle's axes only. No invented numbers. Copy from a verified screenshot later.

Fuel injection (FI) — relative trims

POSITION TH: 20 / 40 / 60 / 80 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	20%	40%	60%	80%
5000				
7500				
10000				
12000				

Ignition (IG) — relative trims

POSITION TH: 15 / 30 / 50 / 70 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	15%	30%	50%	70%
5000				
7500				
10000				
12000				

Cool Morning EMPTY — await Kyle screenshot

Suggested slot: Map 1 · Cold ambient / cool morning — template only until Kyle's cool-weather map exists.

EMPTY on purpose — blank grid with Kyle's axes only. No invented numbers. Copy from a verified screenshot later.

Fuel injection (FI) — relative trims

POSITION TH: 20 / 40 / 60 / 80 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	20%	40%	60%	80%
5000				
7500				
10000				
12000				

Ignition (IG) — relative trims

POSITION TH: 15 / 30 / 50 / 70 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	15%	30%	50%	70%
5000				
7500				
10000				
12000				

Hot Afternoon EMPTY — await Kyle screenshot

Suggested slot: Map 2 · Hot weather / thin air — template only until Kyle's hot-weather map exists.

EMPTY on purpose — blank grid with Kyle's axes only. No invented numbers. Copy from a verified screenshot later.

Fuel injection (FI) — relative trims

POSITION TH: 20 / 40 / 60 / 80 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	20%	40%	60%	80%
5000				
7500				
10000				
12000				

Ignition (IG) — relative trims

POSITION TH: 15 / 30 / 50 / 70 · RPM: 5k / 7k / 10k / 12k

RPM \ TH	15%	30%	50%	70%
5000				
7500				
10000				
12000				

6. Blank FI / IG grid (fill from your phone screenshot)

Copy from Power Tuner or a verified chart. Leave blank if unused. (Print this page twice for Map 1 + Map 2.)

Map name: _____ **Slot:** Map 1 / Map 2

Source: _____ **Date entered:** _____

Fuel injection (FI) — relative trims

Header = TH% (Kyle default 20 / 40 / 60 / 80). Side = RPM (5k / 7.5k / 10k / 12k).

RPM \ TH	20%	40%	60%	80%
5000				
7500				
10000				
12000				

Ignition (IG) — relative trims

Kyle default TH 15 / 30 / 50 / 70 @ same RPMs — confirm on your screenshot.

RPM \ TH	15%	30%	50%	70%
5000				
7500				
10000				
12000				

POSITION / RPM labels used: _____

Notes: _____

7. Maintenance / race log

Date	Hours / motos	Work done	Parts used (PN)	Map pair	Feel / issues	Next

Heat / session notes

Date: _____ Track: _____ Temp / condition: _____

Map used: _____

What worked: _____

What to change next: _____

8. Optional playlists (Spotify)

Leave blank until Spotify drops live URLs — then print or stick in the app.

- **Wrench session:** _____
- **Ride day:** _____
- **Pump-up:** _____

9. Parts quick list (track-day service)

Item	Spec / PN
Oil	Yamalube FS 10W-40 (~1.05 L w/ filter)
Oil filter	5D3-13440-09-00
Filter O-ring	93210-45708
Drain washer	Crush washer (confirm fiche PN)
Air (Powerflow)	Twin Air 152222C kit / Powerflow filters
Air (stock cage)	B2W-14451-01
Spark plug	NGK LMAR9E-J / Yamaha 94701-00424

Affiliate / Amazon links live in the free app Kits page (same price to you; we may earn a commission).

End of Pack — Apple draft body · Janecoon PDF shell · Donny pricing · Yz250f Kits link only · Free app stays ungated